PCBC H Parametric & Sequences Test (Practice C) **Name**_____ _____/16

SHOW ALL WORK FOR FULL CREDIT. ***Non-Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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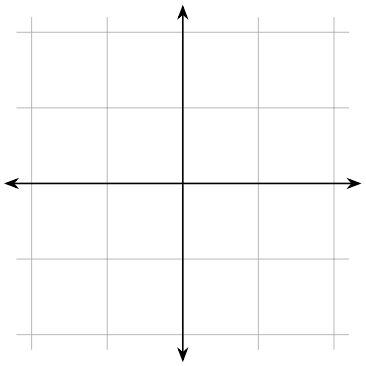
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Signature

Write the pair of parametric equations in rectangular form, including the domain. Then sketch the curve, including direction. (3 points each)

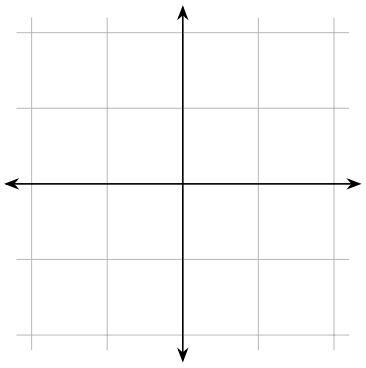
1. $x = 6 - 3t, \quad y = t^2 - 2t, \quad 0 \le t \le 3$

SIMPLIFIED rectangular form: _____



2. $x = 2t, \quad y = \dfrac{4}{t}, \quad 1 \le t \le 4$

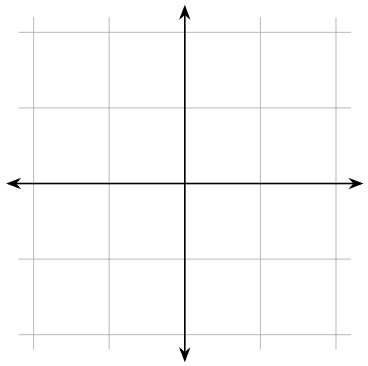
SIMPLIFIED rectangular form: _____



Write the pair of parametric equations in rectangular form. Then sketch the curve. (3 points each)

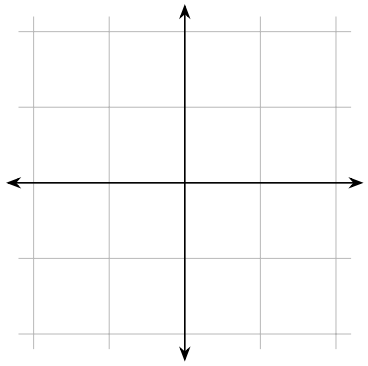
3. $x = 2\cos t, \quad y = -4\sin t, \quad 0 \le t \le 2\pi$

rectangular form: _____



4. $x = -2\sec t, \quad y = 5\tan t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2}$

rectangular form: _____



Convert the rectangular equation to parametric form. Give two different sets of answers. (2 points)

5. $y = -\frac{1}{2}(x-2)^2 + 8$

A sequence is defined recursively. Write the first five terms. (2 points)

6. $a_1 = -3; \quad a_2 = 1; \quad a_n = a_{n-1} + 2a_{n-2}$

Extra practice: the other type #6 could be

Not part of the 16 points. Worked in the key.

Write down the first five terms of the sequence.

E1. $\{a_n\} = \left\{(-1)^{n+1}\dfrac{3^n}{n+2}\right\}$

Answer Key

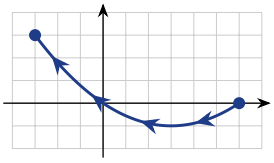
Work on the left, the reason beside it. Boxed = final answer.

Day 1: Non-Calculator

1. $x = 6 - 3t,\ y = t^2 - 2t,\ 0 \le t \le 3$

t	(x,y)
0	$(6,0)$
1	$(3,-1)$
2	$(0,0)$
3	$(-3,3)$

brackets: both ends



starts at $(6,0)$, moves right to left

$t = \frac{6-x}{3} = 2 - \frac{x}{3}$ — *isolate t*

$y = \left(2 - \frac{x}{3}\right)^2 - 2\left(2 - \frac{x}{3}\right)$ — *substitute*

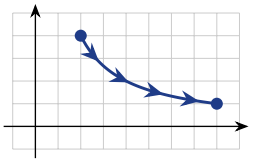
$y = 4 - \frac{4x}{3} + \frac{x^2}{9} - 4 + \frac{2x}{3}$ — *expand*

$\boxed{y = \frac{1}{9}x^2 - \frac{2}{3}x \text{ for } x \text{ in } [-3,6]}$ — *SIMPLIFIED: y alone*

Check: $x = 3$ gives $1 - 2 = -1$. Matches $t = 1$.

2. $x = 2t,\ y = \frac{4}{t},\ 1 \le t \le 4$

t	(x,y)
1	$(2,4)$
2	$(4,2)$
4	$(8,1)$



left to right

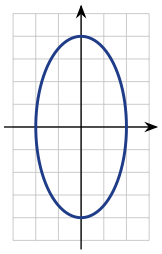
$t = \frac{x}{2}$ — *isolate t*

$y = \dfrac{4}{x/2} = 4 \cdot \frac{2}{x}$ — *dividing by a fraction: flip and multiply*

$\boxed{y = \frac{8}{x} \text{ for } x \text{ in } [2,8]}$ — *only a piece: give the domain*

3. $x = 2\cos t,\ y = -4\sin t,\ 0 \le t \le 2\pi$

t	(x,y)
0	$(2,0)$
$\frac{\pi}{2}$	$(0,-4)$
π	$(-2,0)$
$\frac{3\pi}{2}$	$(0,4)$
2π	$(2,0)$



whole ellipse; direction not asked (clockwise)

$\cos t = \frac{x}{2}, \quad \sin t = -\frac{y}{4}$

$\boxed{\frac{x^2}{4} + \frac{y^2}{16} = 1}$ — *identity; whole ellipse: no domain*

4. $x = -2\sec t,\ y = 5\tan t,\ -\frac{\pi}{2} < t < \frac{\pi}{2}$

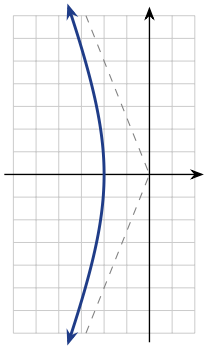
t	(x,y)
$-\frac{\pi}{4}$	$(-2\sqrt{2},-5)$
0	$(-2,0)$
$\frac{\pi}{4}$	$(-2\sqrt{2},5)$

$\sec t > 0$, *so* $x < 0$

$\sec t = -\frac{x}{2}, \quad \tan t = \frac{y}{5}$

$\frac{x^2}{4} - \frac{y^2}{25} = 1$ $\qquad 1 + \tan^2 t = \sec^2 t$

$\boxed{\frac{x^2}{4} - \frac{y^2}{25} = 1 \text{ for } x \text{ in } (-\infty,-2]}$ $\qquad$ *left branch only: a domain cuts it*



left branch; end arrowheads

5. $y = -\frac{1}{2}(x-2)^2 + 8$

$\boxed{x = t,\ y = -\frac{1}{2}(t-2)^2 + 8}$

$x = t+2:\ y = -\frac{1}{2}(t+2-2)^2 + 8$ $\qquad$ *pick* $t+2$ *so the inside is* t

$\boxed{x = t+2,\ y = -\frac{1}{2}t^2 + 8}$

Both for t *in* $(-\infty,\infty)$ *(assumed; writing it is fine). Give both* $x =$ *and* $y =$*; don't expand.*

6. $a_1 = -3;\ a_2 = 1;\ a_n = a_{n-1} + 2a_{n-2}$

$a_3 = a_2 + 2a_1 = 1 + 2(-3) = -5$ $\qquad$ *formula starts at* $n = 3$

$a_4 = a_3 + 2a_2 = -5 + 2(1) = -3$

$a_5 = a_4 + 2a_3 = -3 + 2(-5) = -13$

$\boxed{-3,\ 1,\ -5,\ -3,\ -13}$

Extra practice (not scored)

E1. $\left\{(-1)^{n+1}\frac{3^n}{n+2}\right\}$

$\frac{3}{3},\ -\frac{9}{4},\ \frac{27}{5},\ -\frac{81}{6},\ \frac{243}{7}$ $\qquad$ *plug in* $n = 1$ *to* 5

$\boxed{1,\ -\frac{9}{4},\ \frac{27}{5},\ -\frac{27}{2},\ \frac{243}{7}}$ $\qquad$ *reduce* $\frac{3}{3}$ *and* $\frac{81}{6}$